

Member Contributions — Group 52 (Super Mario Game)

Course: CS202 - C++ Programming / Object-Oriented Design

Class: 25A01

Group: 52

This document is the same data as `Member_Contributions.xlsx` (the TA-facing spreadsheet, in the format used by the group's other CS202 project) rendered as Markdown/PDF for readability — both are generated from `scripts/build_contributions.py` (one source, one task table) so they cannot drift apart.

Summary

Number of students	2
Number of tasks	157
Number of task hours	321
Number of Git commits	480
Max student percentage	0.5
Project score	10

Git commit counts measured at 30f00d8 on 2026-09-02.

Do not edit the grey cells in the spreadsheet edition. TAs enter the score of all PAs in the project score and get the individual scores; students may enter an estimated project score to see how percentages affect it. `Percent` is set to 0.5/0.5 for both members — the raw Tasks/Hours/Git percentages below inform that judgment rather than dictate it, the same convention the reference spreadsheet uses.

No	Student ID	Full name	Tasks	Tasks %	Task Hours	Hours %	Git Commits	Git %	Percent	Score (Student)
1	25125083	Nguyễn Đình Minh Huy	112	71.3%	241	75.1%	419	87.3%	0.5	10
2	25125084	Trần Gia Huy	45	28.7%	80	24.9%	61	12.7%	0.5	10

Note on the gap between the Git % column and the Tasks/Hours % columns: Nguyễn Đình Minh Huy's git identity carries a large share of small fix/doc/log commits (419 of 480) accumulated across the whole session history, while task count and estimated hours — a better proxy for actual design/implementation effort — split closer to 71/29. Both are shown rather than only the more flattering one.

Task-by-task breakdown

A task should be done by only 1-2 students. Each member worked far more than the minimum 5 tasks the reference template asks for.

No	Student ID	Full name	Task Description	Hours	Evidence
1	25125083	Nguyễn Đình Minh Huy	Configured CMake (FetchContent) for SFML 3.0.2 + Dear ImGui/ImGui-SFML and set up the project directory structure	2	see docs/Group52_04/52.md
2	25125083	Nguyễn Đình Minh Huy	Drafted the v2.0 project spec (110 features), UML class diagrams and the implementation plan (TASKS.md, TASK_DIVISION.md)	3	see docs/Group52_04/52.md
3	25125083	Nguyễn Đình Minh Huy	Wrote clean C++ base skeletons for managers, physics and entities	1.5	see docs/Group52_04/52.md
4	25125083	Nguyễn Đình		1	see docs/Group52_04/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy	Implemented MathUtils (vector ops, clamp, interpolation helpers)		
5	25125083	Nguyễn Đình Minh Huy	Built GameStateManager/IGameState state-stack pattern and the Game singleton main loop at a fixed 60Hz timestep	3	see docs/Group52_04/52.md
6	25125083	Nguyễn Đình Minh Huy	Implemented AABB collision primitives and a dynamic SpatialHash broadphase grid	2.5	see docs/Group52_04/52.md
7	25125083	Nguyễn Đình Minh Huy	Built the PhysicsEngine update pipeline (gravity, gravity zones/water viscosity, axis-separated tile resolution)	2.5	see docs/Group52_04/52.md
8	25125083	Nguyễn Đình Minh Huy	Implemented CollisionDetector/ CollisionResolver (solid tiles, conveyors, stomps/damage, player-vs-player bounces)	3	see docs/Group52_04/52.md
9	25125083	Nguyễn Đình Minh Huy	Restructured Entity attributes to protected with friend-class access and read-only getters	1	see docs/Group52_04/52.md
10	25125083	Nguyễn Đình Minh Huy	Implemented the IPlayerState decorator pattern (StarDecorator, MegaDecorator) with timers	2	see docs/Group52_04/52.md
11	25125083	Nguyễn Đình Minh Huy	Coded concrete playable characters (Mario, Luigi, Toad, Peach) with custom physics/abilities	2.5	see docs/Group52_04/52.md
12	25125083	Nguyễn Đình Minh Huy	Built the Item base class and 12 concrete item subclasses (Mushroom, FireFlower, Coin, Star, etc.)	3	see docs/Group52_04/52.md
13	25125084			2	see docs/Group52_04/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Trần Gia Huy	Built InputManager (Command Pattern) with dual-player key mapping (P1 WASD/Space/F, P2 Arrows/M)		
14	25125084	Trần Gia Huy	Implemented SoundManager (Singleton) with music/SFX channels, WAV/OGG loading	1.5	see docs/Group52_04/52.md
15	25125084	Trần Gia Huy	Researched/designed the IMovementStrategy interface and 8 enemy AI strategies	2	see docs/Group52_04/52.md
16	25125084	Trần Gia Huy	Prepared abstract skeletons for the Enemy and Block type hierarchies	1	see docs/Group52_04/52.md
17	25125084	Trần Gia Huy	Designed the EntityFactory registry blueprint for JSON-driven instantiation	1	see docs/Group52_04/52.md
18	25125083	Nguyễn Đình Minh Huy	Implemented the full Camera class (LERP tracking, screen shake, viewport clamping)	2	see docs/Group52_05/52.md
19	25125083	Nguyễn Đình Minh Huy	Refactored move/run commands to intent flags on Character/Player instead of direct velocity changes	1.5	see docs/Group52_05/52.md
20	25125083	Nguyễn Đình Minh Huy	Centralized friction/acceleration/run-speed logic into PhysicsEngine::update() with surface-specific properties	2	see docs/Group52_05/52.md
21	25125083	Nguyễn Đình Minh Huy	Integrated Camera into PlayingState with dual-view (world vs HUD) rendering	1.5	see docs/Group52_05/52.md
22	25125083	Nguyễn Đình Minh Huy	Added mid-air wall-slide friction with a capped slide speed (WALL_SLIDE_SPEED)	1.5	see docs/Group52_05/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
23	25125083	Nguyễn Đình Minh Huy	Integrated nlohmann/json and built LevelLoader to parse tile/entity/spawn data from level JSON	2.5	see docs/Group52_05/52.md
24	25125083	Nguyễn Đình Minh Huy	Authored 4 level files (level_1..3, bonus_1) and an ImGui level selector for playtesting	2	see docs/Group52_05/52.md
25	25125084	Trần Gia Huy	Implemented IMovementStrategy and 8 concrete AI strategies (Patrol, Chase, Fly, etc.)	2.5	see docs/Group52_05/52.md
26	25125084	Trần Gia Huy	Coded the Enemy base class plus Goomba, KoopaTroopa, KoopaParatroopa, Boo	3	see docs/Group52_05/52.md
27	25125084	Trần Gia Huy	Coded the Block base class plus BrickBlock, QuestionBlock, Pipe, Flagpole	2.5	see docs/Group52_05/52.md
28	25125084	Trần Gia Huy	Coded 7 advanced v2.0 enemies (PiranhaPlant, BulletBill, HammerBro, Thwomp, ChainChomp, Lakitu, Spiny)	3.5	see docs/Group52_05/52.md
29	25125084	Trần Gia Huy	Coded 5 advanced v2.0 blocks (HiddenBlock, MovingPlatform, FallingPlatform, IceBlock, ConveyorBelt)	2.5	see docs/Group52_05/52.md
30	25125084	Trần Gia Huy	Implemented EntityFactory to instantiate 25+ entity types from JSON	2.5	see docs/Group52_05/52.md
31	25125084	Trần Gia Huy	Built SpriteSheet, Animation (flyweight) and AnimationManager/Animator	3	see docs/Group52_05/52.md
32	25125084	Trần Gia Huy	Wrote verify_enemies_new.cpp / verify_blocks_new.cpp test harnesses	1.5	see docs/Group52_05/52.md
33	25125083			2.5	see docs/Group52_06/52.md

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		Nguyễn Đình Minh Huy	Built Serializer for save-slot JSON schemas (character, player state, scores, checkpoints, playtime)		
34	25125083	Nguyễn Đình Minh Huy	Implemented StatisticsTracker (EventBus subscriber for coins/enemies/deaths/combos/playtime)	1.5	see docs/Group52_06/52.md
35	25125083	Nguyễn Đình Minh Huy	Implemented AchievementManager (10 achievements) plus a slide-in/out toast HUD	2	see docs/Group52_06/52.md
36	25125083	Nguyễn Đình Minh Huy	Built the ImGui Pause/Settings panel (save slots, volume, keybindings, difficulty, auto-save)	2	see docs/Group52_06/52.md
37	25125083	Nguyễn Đình Minh Huy	Wrote the verify_save_load.cpp integration test suite	1	see docs/Group52_06/52.md
38	25125083	Nguyễn Đình Minh Huy	Implemented the Undo/Redo command framework (EditorCommands)	2	see docs/Group52_06/52.md
39	25125083	Nguyễn Đình Minh Huy	Built the MapEditor viewport dashboard (tile/entity brushes, undo/redo, save)	2	see docs/Group52_06/52.md
40	25125083	Nguyễn Đình Minh Huy	Implemented pixel-to-grid coordinate mapping and grid-line rendering	1.5	see docs/Group52_06/52.md
41	25125083	Nguyễn Đình Minh Huy	Centralized duplicate tile/entity name mappers into SerializationUtils	1	see docs/Group52_06/52.md

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42	25125083	Nguyễn Đình Minh Huy	Wrote the verify_map_editor.cpp test harness	1	see docs/Group52_06/52.md
43	25125084	Trần Gia Huy	Participated in architectural reviews, resolved cross-branch merge conflicts, refined achievement SFX triggers	1	see docs/Group52_06/52.md
44	25125083	Nguyễn Đình Minh Huy	Merged A/feature/save-load and A/feature/level-editor; addressed PR review fixes (EventBus sentinel, debug key remap, Game keybinding encapsulation)	2	see docs/Group52_07/52.md
45	25125083	Nguyễn Đình Minh Huy	Fixed a stale achievement-toast bug on slot reload; aligned achievement keys with SPEC.md	1	see docs/Group52_07/52.md
46	25125083	Nguyễn Đình Minh Huy	Built the consolidated verify_all.cpp test runner combining 6 test suites; updated CMakeLists.txt	2	see docs/Group52_07/52.md
47	25125083	Nguyễn Đình Minh Huy	Authored two_player_ai_plan.md (2-player + Shadow Mario architecture design doc)	2.5	see docs/Group52_07/52.md
48	25125083	Nguyễn Đình Minh Huy	Ran an interactive binary demo to validate save/load and level-editor integration	1	see docs/Group52_07/52.md
49	25125084	Trần Gia Huy	Entity Factory & concrete entities merged into dev (integration of prior work)	0.5	see docs/Group52_07/52.md
50	25125084	Trần Gia Huy	Enemy hierarchy + AI strategies merged into dev (integration of prior work)	0.5	see docs/Group52_07/52.md

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51	25125084	Trần Gia Huy	Blocks/platforms merged into dev (integration of prior work)	0.5	see docs/Group52_07/52.md
52	25125084	Trần Gia Huy	Built Hud (lives/score/coins/time/ stage) plus a boss health bar and retro typography (PressStart2P/ SuperMario256 fonts)	2	see docs/Group52_07/52.md
53	25125084	Trần Gia Huy	Finalized SpriteSheet/ AnimationManager/Animator; added verify_blocks/verify_enemies/ verify_graphics/verify_graphics_visual/ verify_hud_boss test targets	1.5	see docs/Group52_07/52.md
54	25125084	Trần Gia Huy	Consolidated duplicate asset/ and assets/ folders into one assets/ root	0.5	see docs/Group52_07/52.md
55	25125084	Trần Gia Huy	Built the Minimap component (scaled tilemap preview, camera frustum, player tracking dot)	2	Git commit cb6eab2
56	25125084	Trần Gia Huy	Wrote the verify_minimap_visual.cpp test target	1	see docs/Group52_07/52.md
57	25125083	Nguyễn Đình Minh Huy	Reviewed/validated the ParticleSystem/ObjectPool design (O(1) acquire/release, zero-allocation)	1	see docs/Group52_08/52.md
58	25125083	Nguyễn Đình Minh Huy	Updated TASKS.md through Phase 8 and Bonus A, cross-referenced against TASK_DIVISION.md	0.5	see docs/Group52_08/52.md
59	25125083	Nguyễn Đình Minh Huy	Ran git fetch --all, verified dev in sync with origin/dev, tracked branch status	0.5	see docs/Group52_08/52.md
60	25125084			2.5	Git commit 7a6f62b

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		Trần Gia Huy	Implemented ParticleSystem with ObjectPool (burst/continuous emission modes)		
61	25125084	Trần Gia Huy	Tuned particle parameters (alpha fade, emission rate, spread angle)	1.5	Git commits c7d4fd4, 62c78ef
62	25125084	Trần Gia Huy	Built the verify_particles_visual interactive test target	1	see docs/Group52_08/52.md
63	25125084	Trần Gia Huy	Extracted the player sprite sheet player.json (92 animation frames: idle/walk/run/jump/etc.)	3	Git commits d6bd07d, 0be4508, 9403be5
64	25125084	Trần Gia Huy	Built the verify_player_animation_visual test target	1	see docs/Group52_08/52.md
65	25125084	Trần Gia Huy	Began sprite naming standardization / asset pipeline scripts	1	Git commit 15327b4
66	25125084	Trần Gia Huy	Enhanced random-number generation with std::uniform_real_distribution for particle spread	0.5	Git commit 9bb9d16
67	25125083	Nguyễn Đình Minh Huy	Implemented the Fireball entity + ObjectPool, EntityFactory/EventBus wiring, verify_fireball.cpp (4/4 pass)	3	see docs/Group52_09/52.md
68	25125083	Nguyễn Đình Minh Huy	Enhanced Flagpole with height-scaled scoring, animation and a LevelComplete event; verify_flagpole_trampoline.cpp	2	see docs/Group52_09/52.md
69	25125083	Nguyễn Đình Minh Huy	Enhanced Trampoline bounce mechanics (spring compression, configurable impulse)	1.5	see docs/Group52_09/52.md
70	25125083			2	see docs/Group52_09/52.md

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		Nguyễn Đình Minh Huy	Built the CheckpointFlag entity (visual states, auto-save, CheckpointActivated event, respawn coordinates)		
71	25125083	Nguyễn Đình Minh Huy	Implemented the Memento pattern: GameSnapshot + TimeRewindManager (300-frame circular buffer); verify_memento_rewind.cpp (4/4 pass)	3.5	see docs/Group52_09/52.md
72	25125083	Nguyễn Đình Minh Huy	Built MapGenerator for procedural level generation; verify_map_generator.cpp	3	see docs/Group52_09/52.md
73	25125083	Nguyễn Đình Minh Huy	Implemented the dual bounding-box architecture (physicsBox/combatBox) across Entity/CollisionDetector/CollisionResolver	2.5	see docs/Group52_09/52.md
74	25125083	Nguyễn Đình Minh Huy	Added void-fall death detection in PlayingState	1	see docs/Group52_09/52.md
75	25125083	Nguyễn Đình Minh Huy	Added brick/question-block head-butt reactions in PhysicsEngine	1.5	see docs/Group52_09/52.md
76	25125083	Nguyễn Đình Minh Huy	Removed public setLives/setCoins/setScore from Player; added friend class PlayingState	0.5	see docs/Group52_09/52.md
77	25125083	Nguyễn Đình Minh Huy	Added Camera::move() for editor WASD/arrow panning	1	see docs/Group52_09/52.md
78	25125083	Nguyễn Đình	Added a Level Editor Mode (F1) button to MenuState	0.5	see docs/Group52_09/52.md

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		Minh Huy			
79	25125083	Nguyễn Đình Minh Huy	Hardened Game::shutdown() to pop all game states before manager teardown	1	see docs/Group52_09/52.md
80	25125083	Nguyễn Đình Minh Huy	Added multi-path font-loading fallback candidates in PlayingState	0.5	see docs/Group52_09/52.md
81	25125083	Nguyễn Đình Minh Huy	Fetches/merged origin/B/feature/ graphic-visual into dev; resolved font-loading + agent_history.log conflicts; pushed dev	1	Git commit 17cef23
82	25125083	Nguyễn Đình Minh Huy	Fixed non-solid TileType::Coin tiles not being collected on walk-through	1	Git commit d12e763
83	25125083	Nguyễn Đình Minh Huy	Deduplicated a ResourceManager missing-SoundBuffer warning spam	0.5	see docs/Group52_09/52.md
84	25125084	Trần Gia Huy	Built a spritesheet-merging tool; updated enemy_projectile/item/world_scenery_item atlases	3	Git commits 044a5d8, 80c048e, b90
85	25125084	Trần Gia Huy	Cleaned sprite/atlas coordinates affected by the merge	1	Git commit 10828a7
86	25125084	Trần Gia Huy	Updated SPRITE_NAMING_CONVENTIONS.md (frame inventories, naming standards)	1	see docs/Group52_09/52.md
87	25125084	Trần Gia Huy	Built SpriteColorFilter (HSL->RGB, 12Hz Star Power cycling, 15Hz hurt flicker, hit-flash tint)	2	Git commit 2c3bb83

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88	25125084	Trần Gia Huy	Built SpriteTransformAnim (scale-lerp easing, 720 deg/s spin, 12Hz Mega/Mini flicker)	2	Git commit 2c3bb83
89	25125084	Trần Gia Huy	Built the EntityDeathEffect singleton (EnemyFlip, StarKillSpin, PlayerDeathHop)	2	Git commit 2c3bb83
90	25125084	Trần Gia Huy	Added a SoundManager synthesized fallback beep buffer	1	see docs/Group52_09/52.md
91	25125084	Trần Gia Huy	Upgraded verify_graphics_visual with a Transform/Color-Filter FX ImGui tab	1.5	see docs/Group52_09/52.md
92	25125083	Nguyễn Đình Minh Huy	Refined MapGenerator (elevation profiles, biome ceilings, lava pits, jump-reachability-guarded platforms, 3 Star Coins, threat-pacing curve) + ImGui tuning panel	3.5	see docs/Group52_10/52.md
93	25125083	Nguyễn Đình Minh Huy	Built the 3-world campaign + warp-pipe system; generate_game_levels.cpp synthesizing 7 level JSONs; extended SerializationUtils/LevelLoader for warp metadata; seamless sub-level transitions	4	see docs/Group52_10/52.md
94	25125083	Nguyễn Đình Minh Huy	Added ImGui campaign navigation dropdowns and an Active Level Tube warp selector	1.5	see docs/Group52_10/52.md
95	25125083	Nguyễn Đình Minh Huy	Fixed a static-block gravity bug (getGravityMultiplier() override); added item resting-state physics (resolveItemVsBlock)	1.5	see docs/Group52_10/52.md
96	25125083	Nguyễn Đình	Optimized player hitbox dimensions (32px to 24px width tiers); refined stomp-vs-side-hit combat resolution	2	see docs/Group52_10/52.md

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		Minh Huy			
97	25125083	Nguyễn Đình Minh Huy	Fixed the LevelLoader ifstream failbit cascade (file.clear()); expanded fallback search paths	1.5	see docs/Group52_10/52.md
98	25125083	Nguyễn Đình Minh Huy	Fixed a SIGABRT shutdown crash (explicit m_window.close()) and hardened GameStateManager transitions	1.5	see docs/Group52_10/52.md
99	25125084	Trần Gia Huy	Wired sprite animations across all entities; aspect-ratio scaling for HUD/items; POW 8-frame shockwave; trampoline/PSwitch visual states	3.5	Git commits 069817b, 0bc21c3, 2d5058e, 83660b5, 065557d, 61e30
100	25125084	Trần Gia Huy	Built the Web Metadata Editor Studio (metadata_editor.html, edit_sprite_metadata.py) with cursor-anchored zoom, position-finder HUD, drag-to-draw frames, TexturePacker-compatible export	2.5	see docs/Group52_10/52.md
101	25125084	Trần Gia Huy	Integrated sprite HUD assets (animated coin counter, Star Coin indicators, character-specific lives icon)	1.5	see docs/Group52_10/52.md
102	25125084	Trần Gia Huy	Built PipeRenderer (2-block-wide AABB, directional multi-part pipe head/body rendering)	1.5	Git commit a0f1a3b
103	25125084	Trần Gia Huy	Wired game-wide SFX/BGM into SoundManager (jump, stomp, coin, break, power-up/down, boing, stage-clear, game-over) + fallback buffer	2.5	Git commits 1002595, c383c19, 997
104	25125084	Trần Gia Huy	Tuned all 11 enemy AI state machines (Thwomp slam cycle, Boo gaze-detection, HammerBro parabolic throw, Koopa shell physics)	3	Git commit d954a66

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105	25125084	Trần Gia Huy	Built verify_enemies_behavior.cpp + enhanced verify_enemies_visual.cpp (11 isolated test rooms)	1.5	Git commit d954a66
106	25125083	Nguyễn Đình Minh Huy	Resolved the Level 1-3 crash on Bowser's first fireball (mid-frame entity-spawn vector reallocation); rebuilt and playtested on Windows	3	see logs/agent_history.log 2026-08-
107	25125083	Nguyễn Đình Minh Huy	Fixed 12 defects from a playtest pass plus two menus that overflowed	2	Git commit f6fc8bc
108	25125083	Nguyễn Đình Minh Huy	Fixed CMakeLists naming src/Graphics/HUD.cpp when the file is Hud.cpp	0.5	Git commit 03bf828
109	25125083	Nguyễn Đình Minh Huy	Fixed the player being unable to jump while walking into a wall	1	Git commit 60cc735
110	25125083	Nguyễn Đình Minh Huy	Fixed 28 files using std:: symbols without including their header	1	Git commit 4ac8a1c
111	25125083	Nguyễn Đình Minh Huy	Fixed CI building only 2 of the 13 binaries it then tested	1	Git commit 8141120
112	25125083	Nguyễn Đình Minh Huy	Fixed 4 test harnesses opening an sf::RenderWindow on a CI runner with no display	1	Git commit 33d5aae
113	25125083	Nguyễn Đình Minh Huy	Fixed three defects the screenshots showed that the tests did not (backdrop ground line, hill-bottom	2	Git commit da8ca99

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			offset, a Player 2 HUD badge overwriting the world field)		
114	25125083	Nguyễn Đình Minh Huy	Sealed test hermeticity: Serializer::setSaveDirectory() + TestSaveSandbox.hpp so all 23 verify_* harnesses point at a fresh temp directory	2.5	Git commit 045f9b7
115	25125083	Nguyễn Đình Minh Huy	Generated real UML class diagrams from the headers (tools/gen_class_diagram.py) and wrote the CS202 final report's first edition (15 sections, generated by build_report.py)	4	Git commits a0395d6, 3c0a768, 640
116	25125083	Nguyễn Đình Minh Huy	Produced a LaTeX/PDF edition of the report in the HCMUS template and deepened it (implementation section 3->11 subsections, technical-problems section rebuilt into 8 themed groups)	4	Git commits fdec9cc, 730336c
117	25125083	Nguyễn Đình Minh Huy	Diagnosed and fixed three defects reported from a real playthrough: the end-of-level castle's buffer margin, Boom Boom's arena overlapping the 1-2 flagpole, and the camera/z-order bug carried into 1-3; added regression tests for all three	4	this session, branch A/fix/level-completion-and-camera-defects
118	25125083	Nguyễn Đình Minh Huy	Implemented Endless Mode (chunked infinite level generation with rising difficulty, distance-based scoring) and a real solvability oracle for MapGenerator, after evaluating (and rejecting the framework of) the abandoned A/mapgen-gan-plan and A/rl-neural-policy side branches for anything worth porting	5	this session, branch A/feature/endless-mode-and-solvability
119	25125083	Nguyễn Đình Minh Huy	Rebuilt the submission package: a 97-feature list and per-member manual test checklist grounded in a fresh	4	this session, branch A/docs/submission-package-update

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		Minh Huy	source-code audit, the Member Contributions spreadsheet (matching the DesignPatternsGroup52 format), and an expanded report (OOP/pattern rationale, two new UML diagrams)		
120	25125084	Trần Gia Huy	Resolved merge conflicts while pulling the latest changes and merged into main	2	Git commits 72b13c6, a522007, 75e
121	25125083	Nguyễn Đình Minh Huy	Merged all outstanding work to dev/main and generated the initial submission package (AI Usage Declaration, features list, task division, demo-video placeholder, member contributions, final report) while kicking off the defect-auditing test suite.	3	see logs/agent_history.log 2026-08-5 branch main
122	25125083	Nguyễn Đình Minh Huy	Merged the level-completion, endless-mode, and submission-package branches into dev then main and pushed; corrected the AI Usage Declaration's tool attribution (Antigravity vs. Claude Code) per user feedback.	1	see logs/agent_history.log 2026-08-5 09:45 and 10:00, branch main
123	25125083	Nguyễn Đình Minh Huy	Replaced the report's ASCII architecture diagram with a real figure, embedded full-depth UML diagrams in the appendix, and rewrote the class-diagram generator to render every attribute/method with UML visibility notation, fixing a brace-init parsing bug along the way.	3.5	branches A/docs/report-architecture-figure-and-full-uml, A/docs/detailed-full-attributes-methods
124	25125083	Nguyễn Đình Minh Huy	Ran a three-way SPEC/code/report audit via three parallel subagent passes, producing a 15-defect, 14-task (R1-R14) remediation plan and correcting nine stale claims in the features list and report.	4	branch A/docs/spec-feature-audit

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125	25125083	Nguyễn Đình Minh Huy	Small-defect and hygiene batch (R1/R2/R3/R6): fixed SoundManager's level-BGM index mapping, Enemy/Spiny's screen-height despawn bug, and unsurfaced solvability results; deleted the dead AnimationManager class; degraded audio startup gracefully with no device; untracked .member_profile.json and added SPEC's descope addendum.	3.5	see logs/agent_history.log 2026-08-11:48-12:34
126	25125083	Nguyễn Đình Minh Huy	Migrated every raw EventBus SubscriptionId to ScopedSubscription across PlayingState/AchievementManager/StatisticsTracker/Camera (R4), and fixed a real static-destruction-order SIGSEGV the migration surfaced.	2.5	branch A/fix/scoped-subscription-adoption, commit 00c08f8
127	25125083	Nguyễn Đình Minh Huy	Fixed a P-Switch soft-lock caused by mis-themed sub-level floors, a half-rendered single-column pipe in all three sub-levels, and made the overworld QuestionBlock's sprite reflect its emptied state to match SPEC (R16).	3	branch A/fix/sublevel-tile-batch, commit cc6a32d
128	25125083	Nguyễn Đình Minh Huy	Investigated reported 'Bowser never spawns' and 'flag reachable mid-fight' defects (both refuted as already fixed/non-reproducing) and widened Boom Boom's arena from 3.5 to 12.5 tiles of pacing room by removing decorative staircase tiles (R15).	2.5	branch A/fix/boss-encounter-batch, commit 56c8db7
129	25125083	Nguyễn Đình Minh Huy	Fixed the end-game audio batch: castle_complete fanfare being clobbered by level_complete on boss levels via a deferred one-shot-music swap; investigated and struck two further audio defects as not reproducible/already fixed (R19).	2	branch A/fix/endgame-audio-batch, commit de52042

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130	25125083	Nguyễn Đình Minh Huy	Reconciled the SPEC-audit document's defect ledger against shipped code, marking or striking about twenty defects and phases (R1-R19) and documenting four previously mis-recorded playtest defects.	1	branch A/docs/audit-status-reconciliation, commit cc08fc3
131	25125083	Nguyễn Đình Minh Huy	Fixed the Star power's missing rainbow-tint visual, the sub-level Piranha Plant's pipe-centering offset, and the fireball-kill flip-death sprite anchor via a shared Entity::drawSprite fix reaching every flip-killable enemy (R17).	2.5	branch A/fix/player-enemy-visual-ba commit 4550bfe
132	25125083	Nguyễn Đình Minh Huy	Eliminated all 28 real dynamic_casts from CollisionResolver/PhysicsEngine via new virtual Entity/Item/Block/Enemy/Character hooks (nine commits, one dispatch family each), adding 26 regression checks and discovering that Release's -DNDEBUG compiled away nearly all existing assert()-based coverage (R5).	4	branch A/refactor/collision-dispatch, commit 4b87e0d
133	25125083	Nguyễn Đình Minh Huy	Placed all six previously-unused enemy types (KoopParatroopa, Boo, BulletBill, Thwomp, ChainChomp, Lakitu) and hidden blocks across the campaign, fixed the secret_finder achievement's wrong trigger condition, and fixed two playtest-found defects in Bullet Bill and Lakitu placement (R9).	4	branch A/feature/campaign-populati pass, commit 4d6be7e
134	25125083	Nguyễn Đình Minh Huy	Stripped -DNDEBUG from ctest targets only (fixing vacuously-passing assert-based harnesses), refactored the build to compile app sources once via two CMake OBJECT libraries, and added a mutation-tested guard against tests writing into the real saves/ directory (R11).	4	branches A/fix/tests-honour-assertio A/build/object-library, A/fix/hermetic tests

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135	25125083	Nguyễn Đình Minh Huy	Wired eight dormant systems into real gameplay: star-power side-touch kills, four unused particle types, surface-dependent footsteps, combo-hit pitch escalation, menu-row navigation SFX, a single-player camera clamp, and debug-console tab-completion (R7).	3	branch A/feature/wire-dormant-vfx-s commit 7d94fb9
136	25125083	Nguyễn Đình Minh Huy	Added a keyboard-navigable LOAD GAME menu with a 3-slot picker showing character/level/score/time previews, reusing PlayingState's existing loadFromSlot path (R8).	2.5	branch A/feature/load-game-menu, commit d6672b7
137	25125083	Nguyễn Đình Minh Huy	Implemented main-menu Attract Mode (30s idle triggers a bundled demo replay, dismissed by any key), fixing two latent ReplayRecorder playback-pacing/fade-in bugs surfaced by actually watching it run (R10).	3	branch A/feature/attract-mode, commit 6075c27
138	25125083	Nguyễn Đình Minh Huy	Ran a full scripted playtest evidence pass across campaign completion, P-Switch/POW/axe, 2P versus death, difficulty modes, key rebinding, and FPS, discovering two new defects: boss destroyed by the respawn-safety sweep, and Load Game always resuming World 1-1 (R12).	3	branch A/verify/full-playtest-pass
139	25125083	Nguyễn Đình Minh Huy	Fixed the boss-survives-respawn defect (D29), Load Game restoring the saved level (D30), and re-implemented a double-exit guard for PlayingState, each with a mutation-tested regression case; merged and independently re-verified into dev (R20).	3	branch A/fix/remaining-defects-d29-exit, commit 6e4868d
140	25125083	Nguyễn Đình	Root-caused and fixed a user-reported bug where Bowser fell out of the world through his own bridge/lava	2	branch A/fix/boss-falls-out-of-world, commit 95521a8

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		Minh Huy	floor, since the void-plane check only covered players; added an Entity::onLeftLevel virtual hook.		
141	25125083	Nguyễn Đình Minh Huy	Fixed ten gameplay/visual issues in one pass: castle sprite/position, level-clear music timing, Star invincibility flicker, Piranha Plant centering, fireball flip pivot, Bowser arena verification, bonus-level plateau/castle seating, and Peach's Space-key float.	3	see logs/agent_history.log 2026-09-15:31 and 16:52, branch dev
142	25125083	Nguyễn Đình Minh Huy	Fixed sub-level pipe teleports and half-sprite rendering, decoupled the S key from ground-pound (crouch-only), halved Thwomp slam speed with a narrowed detection column, rewrote Chain Chomp's tethered-chase AI, enclosed the 1-3 bridge with walls, and fixed the camera snap on Bowser's defeat.	3	see logs/agent_history.log 2026-09-17:55, branch dev
143	25125083	Nguyễn Đình Minh Huy	Wrote the demo-video-requirements checklist; fixed the M-key minimap toggle, a duplicate player-death animation, metal-footstep SFX volume, and further tuned Thwomp slam/recovery speed and its 1-3 placement.	1.5	see logs/agent_history.log 2026-09-23:30, branch dev
144	25125083	Nguyễn Đình Minh Huy	R21 Phase 1 defect batch: fixed twelve release defects in one pass — a duplicate untracked asset tree causing level-data drift, half/warped pipe rendering, spawns embedded in solid tiles, an incomplete castle slab, double-integrated moving platforms, menu text overflow, duplicate achievement toasts, and gated debug ImGui off by default.	4	branch A/release/defect-batch-r21, commit 171a5b3
145	25125083	Nguyễn Đình Minh Huy	Merged the R21 batch and completed Phases 2-4: built a real GUI level editor wired to EntityFactory (fixing 24	4.5	commit 97fc43a (merge into dev), branch A/release/defect-batch-r21

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		Minh Huy	of 40 silently-broken palette buttons), debug/recording cheats with a void-rescuing Immortal mode, fixed far-chunk procedural entities teleporting back to chunk-local coordinates, an explicit entity-type registry, and Bonus D's dynamic day/night lighting via a real GLSL shader.		
146	25125083	Nguyễn Đình Minh Huy	Second user-reported defect wave: fixed warp-pipe hitbox/art and entry-mode rendering, Bowser's fireball stagger triple-counting a single fireball as three hits, Bowser falling into lava, Lakitu dropping a Fire Flower to keep the fight winnable, the 2P camera freezing on the eliminated player instead of following the survivor, and added save-slot choice/delete.	3.5	commit e97b7da (merge), branch A/versus-camera-and-axes
147	25125083	Nguyễn Đình Minh Huy	Audited OOP/SOLID adherence and design-pattern fidelity across the codebase (dynamic_cast counts, singleton count, friend-class audit, PDF text-overlap root cause) and authored the phased submission-sweep plan with per-lane dependencies, models, and effort levels.	3	docs/issues/ submission_sweep_plan_2026-09-01 branch dev
148	25125083	Nguyễn Đình Minh Huy	Ran the submission-sweep Phase 0 baseline: clean rebuild, full ctest/regression run, a live smoke-test launch, and confirmed the LevelSolvability fix, with no code changes.	1	see logs/agent_history.log 2026-09-01 18:58, branch dev
149	25125083	Nguyễn Đình Minh Huy	Wrote the W11, W12, and W13 weekly reports from git log and agent-history evidence, including tracing the World 1-3 mid-frame entity-spawn crash's root cause for W11.	3	branches A/docs/weekly-w11, A/docs/weekly-w12-w13

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150	25125083	Nguyễn Đình Minh Huy	Archived 20+ stale root/docs artifacts to docs/archive/, untracked stale saves/PDFs/zips, then reverted an AGENTS.md edit that had landed inside AgentHub's generated block and routed the rule change upstream instead.	2	branch A/chore/archive-2026-09-02, commit 9b83c9a
151	25125083	Nguyễn Đình Minh Huy	Extended the AI Usage Declaration to cover the full R1-R21 remediation plan, then corrected its date range and removed an unverifiable test-count claim per review.	1	branch A/docs/ai-declaration, commit 96560bd
152	25125083	Nguyễn Đình Minh Huy	Captured fresh gameplay screenshots and composited three YouTube thumbnail candidates, then narrowed to one final crop-and-zoom version after review found Mario illegibly small at native resolution.	1.5	branch A/docs/video-thumbnail
153	25125083	Nguyễn Đình Minh Huy	Fixed the report PDF's overlapping-table-text defect with content-aware column sizing, added a build.sh Overfull-hbox guard, and split two illegible full-detail UML pages into landscape multi-page spreads.	2.5	branch A/docs/report-layout, commit 0f3e5ea
154	25125083	Nguyễn Đình Minh Huy	Fixed the level editor's F5 playtest routing to Game Over/main menu instead of popping back to the editor on death, with a mutation-tested regression case.	1.5	branch A/fix/editor-playtest-gameover, commit 2bd2f80
155	25125083	Nguyễn Đình Minh Huy	Fixed MapGenerator's procedurally-generated vault exit pipe defaulting to the wrong entry mode; investigated the reported 'unreachable third Hard-mode axe' and found it does not reproduce, shipping a protective regression guard instead of an unjustified data edit.	1.5	branch A/fix/hard-axe-and-vault-pipe

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156	25125083	Nguyễn Đình Minh Huy	Added a camera-distance gate on Endless Mode's ever-growing entity list, added ImGui sliders for the dynamic-lighting tunables, and deleted the dead <code>UiRenderer::wrapText</code> function.	2	branch A/fix/endless-gate-lighting-sliders-wraptext, commit 0fe7138
157	25125083	Nguyễn Đình Minh Huy	Fixed <code>MovingPlatform</code> never advancing its own animation, flipped Spiny's default to spawn as an unhatched egg matching SPEC, and added a hardened post-condition regression test after review found the original test too weak.	2	branch A/fix/moving-platform-and-s-egg, commit cf195cb